

Dataset	$\#F_{total}$	$\min(Var(F))$	$\max(Var(F))$	$\min(F)$	$\max(F)$
Global	4550	0	8286433	0	7689680
$\hookrightarrow$ Lagniez [?]	1948	5	229100	10	399794
$\hookrightarrow\hookrightarrow$ Bayesian Network	1085	32	229100	38	399794
$\hookrightarrow\hookrightarrow$ BMC	18	762	63624	2469	368352
$\hookrightarrow\hookrightarrow$ Circuit	68	26	8704	66	83902
$\hookrightarrow\hookrightarrow$ Configuration	35	1400	2038	1698	11342
$\hookrightarrow\hookrightarrow$ Handmade	68	61	3176	254	174160
$\hookrightarrow\hookrightarrow$ Planning	557	5	24816	10	148891
$\hookrightarrow\hookrightarrow$ QIF	7	251	4473	452	14011
$\hookrightarrow\hookrightarrow$ Random	104	75	250	150	1065
$\hookrightarrow\hookrightarrow$ Scheduling	6	19500	22500	103887	123329
$\hookrightarrow$ Soos [?]	1823	2	8286433	1	7689680
$\hookrightarrow$ Plazar [?]	501	14	486193	31	2598178
$\hookrightarrow\hookrightarrow$ Blasted Real	210	14	10329	35	33008
$\hookrightarrow\hookrightarrow$ Feature Models	133	45	62482	104	343944
$\hookrightarrow\hookrightarrow$ V15	30	17	25615	43	57946
$\hookrightarrow\hookrightarrow$ V3	30	17	25528	31	57586
$\hookrightarrow\hookrightarrow$ V7	30	17	25546	35	57662
$\hookrightarrow\hookrightarrow$ unsorted	68	67	486193	66	2598178
$\hookrightarrow$ Sundermann [?]	278	0	31012	0	350120

Table 1: Dataset summary. The first column indicates the dataset, and the $\#F_{total}$ column indicates how many satisfiable formulae the dataset contains. The following columns indicate the minimum and maximum number of variables (resp. clauses) in the dataset.

Dataset	$\#F_{total}$	$\#D4$	$\#\neg D4$	$\#CapKC \wedge \neg D4$	$\#DivKC \wedge \neg CapKC \wedge \neg D4$
Global	4550	185	670	80	55
$\hookrightarrow$ Lagniez [?]	1948	53	213	40	23
$\hookrightarrow\hookrightarrow$ Bayesian Network	1085	16	70	37	22
$\hookrightarrow\hookrightarrow$ BMC	18	12	2	0	0
$\hookrightarrow\hookrightarrow$ Circuit	68	1	8	0	0
$\hookrightarrow\hookrightarrow$ Configuration	35	1	0	0	0
$\hookrightarrow\hookrightarrow$ Handmade	68	0	32	1	0
$\hookrightarrow\hookrightarrow$ Planning	557	23	92	1	1
$\hookrightarrow\hookrightarrow$ QIF	7	0	1	0	0
$\hookrightarrow\hookrightarrow$ Random	104	0	2	1	0
$\hookrightarrow\hookrightarrow$ Scheduling	6	0	6	0	0
$\hookrightarrow$ Soos [?]	1823	100	393	40	28
$\hookrightarrow$ Plazar [?]	501	31	61	0	4
$\hookrightarrow\hookrightarrow$ Blasted Real	210	6	30	0	3
$\hookrightarrow\hookrightarrow$ Feature Models	133	1	5	0	0
$\hookrightarrow\hookrightarrow$ V15	30	1	5	0	0
$\hookrightarrow\hookrightarrow$ V3	30	1	5	0	0
$\hookrightarrow\hookrightarrow$ V7	30	1	5	0	1
$\hookrightarrow\hookrightarrow$ unsorted	68	21	11	0	0
$\hookrightarrow$ Sundermann [?]	278	1	3	0	0

Table 2: Experimental results regarding the scalability of DivKC and CapKC. Column $\#F_{total}$ indicates the total number of formulae in each dataset. The next column shows the number of formulae compiled only by D4 [?] but not by DivKC or CapKC. Column $\#\neg D4$ shows the number of formulae not compiled by D4. Column $\#CK \wedge \neg D4$ shows the number of formulae that are compiled by CapKC but not by D4. The last column indicates the number of formulae that were only compiled by DivKC, but not by D4 or CapKC.

Dataset	$\#F_{total}$	$\#D4$	$\#\neg D4$	$\#CapKC \wedge \neg D4$	$\#DivKC \wedge \neg CapKC \wedge \neg D4$
Global	4550	163	670	98	40
$\hookrightarrow$ Lagniez [?]	1948	45	213	49	15
$\hookrightarrow\hookrightarrow$ Bayesian Network	1085	12	70	46	14
$\hookrightarrow\hookrightarrow$ BMC	18	12	2	0	0
$\hookrightarrow\hookrightarrow$ Circuit	68	1	8	0	0
$\hookrightarrow\hookrightarrow$ Configuration	35	0	0	0	0
$\hookrightarrow\hookrightarrow$ Handmade	68	0	32	1	0
$\hookrightarrow\hookrightarrow$ Planning	557	20	92	1	1
$\hookrightarrow\hookrightarrow$ QIF	7	0	1	0	0
$\hookrightarrow\hookrightarrow$ Random	104	0	2	1	0
$\hookrightarrow\hookrightarrow$ Scheduling	6	0	6	0	0
$\hookrightarrow$ Soos [?]	1823	91	393	49	21
$\hookrightarrow$ Plazar [?]	501	26	61	0	4
$\hookrightarrow\hookrightarrow$ Blasted Real	210	6	30	0	3
$\hookrightarrow\hookrightarrow$ Feature Models	133	1	5	0	0
$\hookrightarrow\hookrightarrow$ V15	30	0	5	0	0
$\hookrightarrow\hookrightarrow$ V3	30	1	5	0	0
$\hookrightarrow\hookrightarrow$ V7	30	0	5	0	1
$\hookrightarrow\hookrightarrow$ unsorted	68	18	11	0	0
$\hookrightarrow$ Sundermann [?]	278	1	3	0	0

Table 3: Experimental results regarding the scalability of DivKC and CapKC. Column $\#F_{total}$ indicates the total number of formulae in each dataset. The next column shows the number of formulae compiled only by D4 [?] but not by DivKC or CapKC. Column $\#\neg D4$ shows the number of formulae not compiled by D4. Column $\#CK \wedge \neg D4$ shows the number of formulae that are compiled by CapKC but not by D4. The last column indicates the number of formulae that were only compiled by DivKC, but not by D4 or CapKC. In this experiment, CapKC was limited to 8GB and 360s for each D4 trial.

Dataset	$\#F_{total}$	$\#\neg D4$	$\#\neg D4 \wedge CapKC$	$\#\neg D4 \wedge DivKC$	$\#CapKC \wedge \neg DivKC \wedge \neg D4$
Global	4550	670	80	114	21
$\hookrightarrow$ Lagniez [?]	1948	213	40	52	11
$\hookrightarrow\hookrightarrow$ Bayesian Network	1085	70	37	51	8
$\hookrightarrow\hookrightarrow$ BMC	18	2	0	0	0
$\hookrightarrow\hookrightarrow$ Circuit	68	8	0	0	0
$\hookrightarrow\hookrightarrow$ Configuration	35	0	0	0	0
$\hookrightarrow\hookrightarrow$ Handmade	68	32	1	0	1
$\hookrightarrow\hookrightarrow$ Planning	557	92	1	1	1
$\hookrightarrow\hookrightarrow$ QIF	7	1	0	0	0
$\hookrightarrow\hookrightarrow$ Random	104	2	1	0	1
$\hookrightarrow\hookrightarrow$ Scheduling	6	6	0	0	0
$\hookrightarrow$ Soos [?]	1823	393	40	58	10
$\hookrightarrow$ Plazar [?]	501	61	0	4	0
$\hookrightarrow\hookrightarrow$ Blasted Real	210	30	0	3	0
$\hookrightarrow\hookrightarrow$ Feature Models	133	5	0	0	0
$\hookrightarrow\hookrightarrow$ V15	30	5	0	0	0
$\hookrightarrow\hookrightarrow$ V3	30	5	0	0	0
$\hookrightarrow\hookrightarrow$ V7	30	5	0	1	0
$\hookrightarrow\hookrightarrow$ unsorted	68	11	0	0	0
$\hookrightarrow$ Sundermann [?]	278	3	0	0	0

Table 4: Experimental results regarding the scalability of DivKC and CapKC.

Dataset	$\#F_{total}$	$\#\neg D4$	$\#\neg D4 \wedge CapKC$	$\#\neg D4 \wedge DivKC$	$\#CapKC \wedge \neg DivKC \wedge \neg D4$
Global	4550	670	98	114	24
$\hookrightarrow$ Lagniez [?]	1948	213	49	52	12
$\hookrightarrow\hookrightarrow$ Bayesian Network	1085	70	46	51	9
$\hookrightarrow\hookrightarrow$ BMC	18	2	0	0	0
$\hookrightarrow\hookrightarrow$ Circuit	68	8	0	0	0
$\hookrightarrow\hookrightarrow$ Configuration	35	0	0	0	0
$\hookrightarrow\hookrightarrow$ Handmade	68	32	1	0	1
$\hookrightarrow\hookrightarrow$ Planning	557	92	1	1	1
$\hookrightarrow\hookrightarrow$ QIF	7	1	0	0	0
$\hookrightarrow\hookrightarrow$ Random	104	2	1	0	1
$\hookrightarrow\hookrightarrow$ Scheduling	6	6	0	0	0
$\hookrightarrow$ Soos [?]	1823	393	49	58	12
$\hookrightarrow$ Plazar [?]	501	61	0	4	0
$\hookrightarrow\hookrightarrow$ Blasted Real	210	30	0	3	0
$\hookrightarrow\hookrightarrow$ Feature Models	133	5	0	0	0
$\hookrightarrow\hookrightarrow$ V15	30	5	0	0	0
$\hookrightarrow\hookrightarrow$ V3	30	5	0	0	0
$\hookrightarrow\hookrightarrow$ V7	30	5	0	1	0
$\hookrightarrow\hookrightarrow$ unsorted	68	11	0	0	0
$\hookrightarrow$ Sundermann [?]	278	3	0	0	0

Table 5: Experimental results regarding the scalability of DivKC and CapKC. In this experiment, CapKC was limited to 8GB and 360s for each D4 trial.

Dataset	$\#F$	$Y_l \leq R_F $	$Y_h \geq R_F $	Coverage	$ R_{GP} \leq R_F \leq R_{GU} $
Global	3670	0.969	0.857	0.826	1.000
$\hookrightarrow$ Lagniez [?]	1689	0.969	0.895	0.864	1.000
$\hookrightarrow\hookrightarrow$ Bayesian Network	1009	0.985	0.908	0.893	1.000
$\hookrightarrow\hookrightarrow$ BMC	4	1.000	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ Circuit	59	1.000	0.814	0.814	1.000
$\hookrightarrow\hookrightarrow$ Configuration	33	0.788	0.697	0.485	1.000
$\hookrightarrow\hookrightarrow$ Handmade	35	1.000	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ Planning	441	0.932	0.862	0.794	1.000
$\hookrightarrow\hookrightarrow$ QIF	6	1.000	0.833	0.833	1.000
$\hookrightarrow\hookrightarrow$ Random	102	0.990	0.990	0.980	1.000
$\hookrightarrow\hookrightarrow$ Scheduling	0				
$\hookrightarrow$ Soos [?]	1307	0.969	0.906	0.875	1.000
$\hookrightarrow$ Plazar [?]	403	0.968	0.789	0.757	1.000
$\hookrightarrow\hookrightarrow$ Blasted Real	176	1.000	0.938	0.938	1.000
$\hookrightarrow\hookrightarrow$ Feature Models	124	0.968	0.460	0.427	1.000
$\hookrightarrow\hookrightarrow$ V15	23	1.000	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ V3	23	1.000	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ V7	23	1.000	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ unsorted	34	0.735	0.794	0.529	1.000
$\hookrightarrow$ Sundermann [?]	271	0.967	0.487	0.454	1.000

Table 6: Experimental results for DivKC-AMC. Column $\#F$ indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	#F	$Y_l \leq R_F $	$Y_h \geq R_F $	Coverage	$ R_F \leq R_{G_U} $
Global	323	0.994	0.988	0.981	1.000
$\hookrightarrow$ Lagniez [?]	168	0.994	0.988	0.982	1.000
$\hookrightarrow \hookrightarrow$ Bayesian Network	138	0.993	0.993	0.986	1.000
$\hookrightarrow \hookrightarrow$ BMC	0				
$\hookrightarrow \hookrightarrow$ Circuit	2	1.000	1.000	1.000	1.000
$\hookrightarrow \hookrightarrow$ Configuration	8	1.000	0.875	0.875	1.000
$\hookrightarrow \hookrightarrow$ Handmade	1	1.000	1.000	1.000	1.000
$\hookrightarrow \hookrightarrow$ Planning	17	1.000	1.000	1.000	1.000
$\hookrightarrow \hookrightarrow$ QIF	1	1.000	1.000	1.000	1.000
$\hookrightarrow \hookrightarrow$ Random	1	1.000	1.000	1.000	1.000
$\hookrightarrow \hookrightarrow$ Scheduling	0				
$\hookrightarrow$ Soos [?]	149	1.000	0.987	0.987	1.000
$\hookrightarrow$ Plazar [?]	4	1.000	1.000	1.000	1.000
$\hookrightarrow \hookrightarrow$ Blasted Real	3	1.000	1.000	1.000	1.000
$\hookrightarrow \hookrightarrow$ Feature Models	0				
$\hookrightarrow \hookrightarrow$ V15	0				
$\hookrightarrow \hookrightarrow$ V3	0				
$\hookrightarrow \hookrightarrow$ V7	0				
$\hookrightarrow \hookrightarrow$ unsorted	1	1.000	1.000	1.000	1.000
$\hookrightarrow$ Sundermann [?]	2	0.500	1.000	0.500	1.000

Table 7: Experimental results for CapKC. Column #F indicates with how many formulae the statistics have been computed. The 'Coverage' column indicates how often $|R_F|$ is within the confidence interval $[Y_l; Y_h]$ and thus measures the accuracy of our method.

Dataset	#F	$Y_l \geq R_{G_P} $	$Y_h \leq R_{G_U} $	Both	$median(r_c)$	$max(r_c)$
Global	3669	0.807	0.836	0.721	4.17e-9	0.54
$\hookrightarrow$ Lagniez [?]	1689	0.844	0.869	0.760	1.8e-9	0.54
$\hookrightarrow \hookrightarrow$ Bayesian Network	1009	0.935	0.899	0.850	8.16e-7	0.0963
$\hookrightarrow \hookrightarrow$ BMC	4	1.000	1.000	1.000	0.0	2.34e-5
$\hookrightarrow \hookrightarrow$ Circuit	59	0.932	0.814	0.797	5.0e-30	4.27e-7
$\hookrightarrow \hookrightarrow$ Configuration	33	0.394	0.485	0.303	4.54e-6	0.54
$\hookrightarrow \hookrightarrow$ Handmade	35	1.000	1.000	1.000	0.0	0.0
$\hookrightarrow \hookrightarrow$ Planning	441	0.834	0.798	0.735	7.59e-40	0.0181
$\hookrightarrow \hookrightarrow$ QIF	6	0.667	0.833	0.500	1.12e-56	1.89e-5
$\hookrightarrow \hookrightarrow$ Random	102	0.029	0.990	0.029	1.84e-68	1.92e-23
$\hookrightarrow \hookrightarrow$ Scheduling	0					
$\hookrightarrow$ Soos [?]	1307	0.920	0.893	0.834	1.75e-7	0.0962
$\hookrightarrow$ Plazar [?]	403	0.697	0.769	0.610	1.64e-13	0.0153
$\hookrightarrow \hookrightarrow$ Blasted Real	176	0.932	0.938	0.881	2.99e-13	0.0153
$\hookrightarrow \hookrightarrow$ Feature Models	124	0.202	0.427	0.048	3.8e-13	0.00975
$\hookrightarrow \hookrightarrow$ V15	23	1.000	1.000	1.000	2.51e-17	0.00603
$\hookrightarrow \hookrightarrow$ V3	23	1.000	1.000	1.000	2.92e-17	0.00401
$\hookrightarrow \hookrightarrow$ V7	23	0.957	1.000	0.957	2.35e-17	0.00406
$\hookrightarrow \hookrightarrow$ unsorted	34	0.706	0.676	0.500	7.57e-91	2.39e-9
$\hookrightarrow$ Sundermann [?]	270	0.196	0.448	0.093	1.69e-13	0.281

Table 8: Experimental results comparing the bounds obtained with DivKC-AMC and with Lemma ???. Column #F indicates with how many formulae the statistics have been computed. The 'Both' column indicates how often we have $Y_l \geq |R_{G_P}| \wedge Y_l \leq |R_F|$ and $Y_h \leq |R_{G_U}| \wedge Y_h \geq |R_F|$. The last two columns represent the observed median and maximum values of the ratio $r_c = \frac{\min(Y_h, |R_{G_U}|) - \max(Y_l, |R_{G_P}|)}{|R_{G_U}| - |R_{G_P}|}$, which was calculated exclusively if $Y_l \leq |R_F| \leq Y_h$. The number of formulae on which the last two columns are computed can easily be obtained by multiplying the #F column with the 'Coverage' column in Table 6.

Dataset	#F	$Y_h \leq R_{G_U} $	$median(r_c)$	$max(r_c)$
Global	334	0.994	0.00794	0.0257
$\hookrightarrow$ Lagniez [?]	175	0.994	0.0072	0.0257
$\hookrightarrow\hookrightarrow$ Bayesian Network	143	1.000	0.00915	0.0257
$\hookrightarrow\hookrightarrow$ BMC	0			
$\hookrightarrow\hookrightarrow$ Circuit	2	1.000	0.00757	0.0103
$\hookrightarrow\hookrightarrow$ Configuration	8	0.875	0.00654	0.0256
$\hookrightarrow\hookrightarrow$ Handmade	2	1.000	0.0135	0.0231
$\hookrightarrow\hookrightarrow$ Planning	18	1.000	0.00164	0.0257
$\hookrightarrow\hookrightarrow$ QIF	1	1.000	0.0257	0.0257
$\hookrightarrow\hookrightarrow$ Random	1	1.000	0.00422	0.00422
$\hookrightarrow\hookrightarrow$ Scheduling	0			
$\hookrightarrow$ Soos [?]	153	0.993	0.00837	0.0257
$\hookrightarrow$ Plazar [?]	4	1.000	0.00684	0.0257
$\hookrightarrow\hookrightarrow$ Blasted Real	3	1.000	0.00339	0.0257
$\hookrightarrow\hookrightarrow$ Feature Models	0			
$\hookrightarrow\hookrightarrow$ V15	0			
$\hookrightarrow\hookrightarrow$ V3	0			
$\hookrightarrow\hookrightarrow$ V7	0			
$\hookrightarrow\hookrightarrow$ unsorted	1	1.000	0.0103	0.0103
$\hookrightarrow$ Sundermann [?]	2	1.000	0.00981	0.0155

Table 9: Experimental results comparing the bounds obtained with CapKC. Column #F indicates with how many formulae the statistics have been computed. The last two columns represent the observed median and maximum values of the relative confidence-interval width $r_c = \frac{Y_h - Y_l}{|R_{G_U}|}$.

Dataset	#F	$l \leq Y_{ApproxMC} \leq h$	$l \leq Y_{DivKC-AMC} \leq h$	$l \leq Y_{CapKC-AMC} \leq h$
Global	2814	0.939	0.875	0.982
$\hookrightarrow$ Lagniez [?]	1353	0.982	0.891	0.981
$\hookrightarrow\hookrightarrow$ Bayesian Network	922	1.000	0.851	0.975
$\hookrightarrow\hookrightarrow$ BMC	3	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ Circuit	48	0.917	0.875	1.000
$\hookrightarrow\hookrightarrow$ Configuration	16	1.000	0.875	1.000
$\hookrightarrow\hookrightarrow$ Handmade	32	0.375	1.000	1.000
$\hookrightarrow\hookrightarrow$ Planning	319	1.000	0.994	0.991
$\hookrightarrow\hookrightarrow$ QIF	5	1.000	0.800	1.000
$\hookrightarrow\hookrightarrow$ Random	8	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ Scheduling	0			
$\hookrightarrow$ Soos [?]	1186	0.957	0.848	0.979
$\hookrightarrow$ Plazar [?]	253	0.613	0.905	0.996
$\hookrightarrow\hookrightarrow$ Blasted Real	149	1.000	0.886	0.993
$\hookrightarrow\hookrightarrow$ Feature Models	6	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ V15	23	0.000	0.913	1.000
$\hookrightarrow\hookrightarrow$ V3	23	0.000	0.957	1.000
$\hookrightarrow\hookrightarrow$ V7	22	0.000	0.955	1.000
$\hookrightarrow\hookrightarrow$ unsorted	30	0.000	0.900	1.000
$\hookrightarrow$ Sundermann [?]	22	1.000	0.955	1.000

Table 10: Experimental results comparing the accuracy of DivKC-AMC and CapKC-AMC with ApproxMC 7. Column #F indicates with how many formulae the statistics have been computed. Column $l \leq Y_A \leq h$ indicates how often the model count returned by A is within the indicated bounds, with $l = \frac{|R_F|}{1.2}$ and $h = 1.2 \times |R_F|$.

Dataset	#F	$l \leq Y_{ApproxMC} \leq h$	$l \leq Y_{DivKC-AMC} \leq h$	$l \leq Y_{CapKC-AMC} \leq h$
Global	298	0.997	0.500	0.826
$\hookrightarrow$ Lagniez [?]	148	1.000	0.486	0.824
$\hookrightarrow\hookrightarrow$ Bayesian Network	138	1.000	0.449	0.833
$\hookrightarrow\hookrightarrow$ BMC	0			
$\hookrightarrow\hookrightarrow$ Circuit	2	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ Configuration	0			
$\hookrightarrow\hookrightarrow$ Handmade	0			
$\hookrightarrow\hookrightarrow$ Planning	7	1.000	1.000	0.571
$\hookrightarrow\hookrightarrow$ QIF	1	1.000	1.000	1.000
$\hookrightarrow\hookrightarrow$ Random	0			
$\hookrightarrow\hookrightarrow$ Scheduling	0			
$\hookrightarrow$ Soos [?]	147	0.993	0.503	0.830
$\hookrightarrow$ Plazar [?]	3	1.000	1.000	0.667
$\hookrightarrow\hookrightarrow$ Blasted Real	3	1.000	1.000	0.667
$\hookrightarrow\hookrightarrow$ Feature Models	0			
$\hookrightarrow\hookrightarrow$ V15	0			
$\hookrightarrow\hookrightarrow$ V3	0			
$\hookrightarrow\hookrightarrow$ V7	0			
$\hookrightarrow\hookrightarrow$ unsorted	0			
$\hookrightarrow$ Sundermann [?]	0			

Table 11: Experimental results comparing the accuracy of DivKC-AMC and CapKC-AMC with ApproxMC 7. Column #F indicates with how many formulae the statistics have been computed. Column $l \leq Y_A \leq h$ indicates how often the model count returned by A is within the indicated bounds, with $l = \frac{|R_F|}{1.2}$ and $h = 1.2 \times |R_F|$. We only consider formulae for which $|R_{G_U}| \neq |R_F|$, thereby excluding cases where $CapKC$ returns $G_U = F$ and the model count can be obtained directly from G_U .

Dataset	#F	#DivKC-AMC	$\log_{10}(min)$	$mean$	$median$	$\log_{10}(max)$
Global	3291	2122	-5.4	153.0	1.8	5.1
$\hookrightarrow$ Lagniez [?]	1623	1156	-2.9	224.2	2.1	4.9
$\hookrightarrow\hookrightarrow$ Bayesian Network	1051	705	-2.7	8.1	1.9	3.8
$\hookrightarrow\hookrightarrow$ BMC	4	0	-1.5	0.4	0.4	-0.2
$\hookrightarrow\hookrightarrow$ Circuit	56	28	-2.9	1.6	1.0	1.2
$\hookrightarrow\hookrightarrow$ Configuration	16	16	0.5	5251.7	50.5	4.9
$\hookrightarrow\hookrightarrow$ Handmade	35	14	-1.8	95.3	0.1	3.4
$\hookrightarrow\hookrightarrow$ Planning	352	287	-2.9	729.0	4.6	4.6
$\hookrightarrow\hookrightarrow$ QIF	7	5	-0.6	2.4	2.9	0.6
$\hookrightarrow\hookrightarrow$ Random	102	101	-0.4	111.6	36.5	2.6
$\hookrightarrow\hookrightarrow$ Scheduling	0					
$\hookrightarrow$ Soos [?]	1358	851	-3.5	6.2	1.7	3.6
$\hookrightarrow$ Plazar [?]	288	98	-5.4	1.0	0.3	1.2
$\hookrightarrow\hookrightarrow$ Blasted Real	177	86	-3.2	1.2	0.9	0.7
$\hookrightarrow\hookrightarrow$ Feature Models	6	4	-0.4	4.7	3.1	1.2
$\hookrightarrow\hookrightarrow$ V15	23	0	-3.3	0.0	0.0	-1.2
$\hookrightarrow\hookrightarrow$ V3	24	0	-5.4	0.0	0.0	-1.2
$\hookrightarrow\hookrightarrow$ V7	23	0	-3.0	0.0	0.0	-1.1
$\hookrightarrow\hookrightarrow$ unsorted	35	8	-3.6	1.1	0.3	1.0
$\hookrightarrow$ Sundermann [?]	22	17	-1.2	5954.6	12.6	5.1

Table 12: Experimental results comparing the runtime of DivKC-AMC with ApproxMC 7 for 20 runs. Column #F indicates the number of formulae over which the statistics were computed, and column #DivKC-AMC shows how often DivKC-AMC was faster than ApproxMC 7. The remaining columns report how much faster DivKC-AMC was, based on the logarithm of the minimum, the mean, the median, and the logarithm of the maximum of the ratio: ApproxMC 7 execution time divided by DivKC-AMC execution time.

Dataset	#F	#CapKC-AMC	$\log_{10}(\min)$	$mean$	$median$	$\log_{10}(\max)$
Global	2838	1629	-3.5	162.7	1.6	5.0
$\hookrightarrow$ Lagniez [?]	1352	820	-2.6	251.6	1.8	4.7
$\hookrightarrow\hookrightarrow$ Bayesian Network	924	512	-2.1	17.9	1.4	4.2
$\hookrightarrow\hookrightarrow$ BMC	2	1	-0.8	1.4	1.4	0.4
$\hookrightarrow\hookrightarrow$ Circuit	47	28	-1.8	2.0	1.7	1.0
$\hookrightarrow\hookrightarrow$ Configuration	16	16	0.5	3947.4	377.9	4.7
$\hookrightarrow\hookrightarrow$ Handmade	33	9	-2.6	2.0	0.0	1.3
$\hookrightarrow\hookrightarrow$ Planning	318	248	-1.3	818.6	4.9	4.7
$\hookrightarrow\hookrightarrow$ QIF	5	2	-0.9	1.4	0.2	0.6
$\hookrightarrow\hookrightarrow$ Random	7	4	-2.0	1.1	1.0	0.4
$\hookrightarrow\hookrightarrow$ Scheduling	0					
$\hookrightarrow$ Soos [?]	1204	679	-2.1	6.1	1.5	3.2
$\hookrightarrow$ Plazar [?]	259	111	-3.5	1.5	0.7	1.2
$\hookrightarrow\hookrightarrow$ Blasted Real	148	91	-1.2	2.2	1.7	1.2
$\hookrightarrow\hookrightarrow$ Feature Models	6	5	-0.5	4.9	3.6	1.1
$\hookrightarrow\hookrightarrow$ V15	24	0	-3.5	0.0	0.0	-1.3
$\hookrightarrow\hookrightarrow$ V3	24	0	-2.8	0.0	0.0	-1.3
$\hookrightarrow\hookrightarrow$ V7	24	0	-3.2	0.0	0.0	-1.0
$\hookrightarrow\hookrightarrow$ unsorted	33	15	-1.1	1.5	0.9	0.8
$\hookrightarrow$ Sundermann [?]	23	19	-1.2	4948.0	10.1	5.0

Table 13: Experimental results comparing the runtime of CapKC-AMC with ApproxMC 7 for 20 runs. Column #F indicates the number of formulae over which the statistics were computed, and column #CapKC-AMC shows how often CapKC-AMC was faster than ApproxMC 7. The remaining columns report how much faster CapKC-AMC was, based on the logarithm of the minimum, the mean, the median, and the logarithm of the maximum of the ratio: ApproxMC 7 execution time divided by CapKC-AMC execution time.

Dataset	#F	#CapKC-AMC	$\log_{10}(\min)$	$mean$	$median$	$\log_{10}(\max)$
Global	294	2	-2.1	1.9	0.0	2.7
$\hookrightarrow$ Lagniez [?]	143	2	-2.1	3.8	0.0	2.7
$\hookrightarrow\hookrightarrow$ Bayesian Network	136	0	-2.1	0.2	0.0	-0.1
$\hookrightarrow\hookrightarrow$ BMC	0					
$\hookrightarrow\hookrightarrow$ Circuit	1	0	-1.8	0.0	0.0	-1.8
$\hookrightarrow\hookrightarrow$ Configuration	0					
$\hookrightarrow\hookrightarrow$ Handmade	0					
$\hookrightarrow\hookrightarrow$ Planning	5	2	-1.3	105.0	0.1	2.7
$\hookrightarrow\hookrightarrow$ QIF	1	0	-0.8	0.2	0.2	-0.8
$\hookrightarrow\hookrightarrow$ Random	0					
$\hookrightarrow\hookrightarrow$ Scheduling	0					
$\hookrightarrow$ Soos [?]	149	0	-2.1	0.2	0.0	-0.1
$\hookrightarrow$ Plazar [?]	1	0	-1.2	0.1	0.1	-1.2
$\hookrightarrow\hookrightarrow$ Blasted Real	1	0	-1.2	0.1	0.1	-1.2
$\hookrightarrow\hookrightarrow$ Feature Models	0					
$\hookrightarrow\hookrightarrow$ V15	0					
$\hookrightarrow\hookrightarrow$ V3	0					
$\hookrightarrow\hookrightarrow$ V7	0					
$\hookrightarrow\hookrightarrow$ unsorted	0					
$\hookrightarrow$ Sundermann [?]	1	0	-0.3	0.5	0.5	-0.3

Table 14: CapKC vs Approxmc7. In the cases where, for CapKC, we have $|R_F| \neq |R_{G_U}|$.